Run Book for ViewCompare

Date

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Introduction

Denodo periodically releases both major and minor updates to its platform. During migrations—whether between major versions or incremental updates—customers are required to perform thorough testing to ensure consistency and reliability across environments. To facilitate and accelerate this process, the ViewCompare has been developed to identify differences in both metadata and data across Denodo environments.

The framework enables comparison of view definitions (metadata) as well as the underlying data, helping users detect discrepancies that may arise during migration. It is designed to provide detailed insights, including **record-level differences**, for a specified set of views across different versions.

In addition to detailed comparisons, the framework offers flexibility by allowing users to perform basic validation checks or opt for more granular, record-level analysis based on their requirements. This adaptability ensures that the framework can cater to varying levels of testing depth.

The ViewCompare can be deployed on any Denodo instance and configured to connect to both source and target environments. This allows users to perform comprehensive validation and View Validation Testing across different systems in a centralized and efficient manner.

Scope

The **ViewCompare** is designed to assist customers in accelerating the testing process (limited to comparing the views) during migrations to newer Denodo versions or when applying platform updates. It provides a foundational setup that enables efficient comparison of views across Denodo environments, whether within the same server or across different servers.

The framework supports the following types of comparisons between two Denodo environments:

1. **Metadata Comparison**

   Evaluates differences in view definitions, including schema structure, field names, data types, and more.

2. **Row Count Comparison**

   Compares the total number of records returned by each view to identify any discrepancies in data volume.

3. **Row-Level Data Comparison**

   Performs detailed comparison of the actual data returned by the views. Helps identify mismatches at the record level, enabling precise validation of data consistency.

Key Configuration Concepts

The Denodo ViewCompare solution must be configured within a Denodo V9 instance. The overall setup is organized into four key components (or "tiles"), each representing a specific role in the framework:

1. **Host VDP server**
   - The Denodo server where the ViewCompare is deployed and executed.
   - This instance orchestrates the comparison process between environments.
   - The host must be a Denodo V9 server and it is recommended that the host be a non-production server, such as a Sandbox, to ensure minimal operational impact. The dedicated Host server connects to both the **Source** and **Target** VDP servers.
2. **Source VDP server**
   - Represents the existing or current Denodo environment.
   - This serves as the baseline against which the target environment is validated.
3. **Target VDP server**
   - Refers to the new or upgraded Denodo environment.
   - This environment is compared against the Source to identify any discrepancies.
4. **Report Tables**
   - A relational database (RDBMS) used to store the results generated by the ViewCompare.
   - The stored results remain available until the next execution cycle.
   - During each execution, existing data is cleared using a truncate operation and reloaded with the latest results. This approach ensures that only the most recent execution data is retained.

Prerequisites

The following prerequisites must be fulfilled before setting up the framework:

1. **Relational Database**
   - A relational database must be available to store the output generated by the ViewCompare.
   - Ensure that the database is accessible from the environment where the framework will be executed.

2. **Python Environment**
   - Python version **3.10 or later** must be installed on the machine where the framework is being configured (Host VDP server).
   - Additionally, the required Python library **JayDeBeApi** must be installed to enable JDBC connectivity from the Python script.

3. **Denodo Connect Component**
   - The **SSH Custom Wrapper** (Denodo Connect component) must be properly configured in the Denodo instance where the framework will be deployed(Host VDP Server).
   - This component is required to facilitate remote execution of scripts via SSH.

4. **SSH Server Setup**
   - An SSH server must be installed and running on the machine where the Python script is hosted (Host VDP server).
   - Ensure that the server is properly configured to accept incoming connections from the Denodo environment.

5. **Port and Connectivity:**
   - For the ViewCompare to successfully communicate with both the Source and Target VDP servers, it is essential that these servers are reachable from the Host VDP server over the network. Specifically, the Host VDP server must have proper connectivity to the **Source VDP server** and **Target VDP server** and port 9999 (the default VDP port) should be open and accessible on both ends. The SSH Server port number (default is 22) should also be open and accessible from the Host VDP server.

6. **Primary key for Row-Level Data Comparison:**
   - For the Row-Level Data Comparison to function accurately, it is essential that the views being compared contain a column that uniquely identifies each record. This unique identifier ensures precise matching of rows between the Source and Target datasets during comparison. While this field does not need to be explicitly defined as a primary key within Denodo, it must consistently contain unique, non-null values across the dataset to avoid mismatches or incorrect comparison results.

Setup and Configuration of ViewCompare

The Setup of ViewCompare is done in the Host VDP Server. Follow the steps outlined in each of the sections below.

Report Table

To enable persistence of testing results, the required report tables must be created in a relational database. Please follow the steps below:

1. **Select a Supported Database**
   - Choose any JDBC-supported relational database where the testing results will be stored.
   - Ensure that the database is accessible from the Denodo environment (Host VDP server).

2. **Execute the Appropriate database DDL Script**
   - Navigate to the **<folder_name>/DDL** folder containing DDL scripts for multiple database systems.
   - Select the script corresponding to your chosen database (e.g., SQL Server, Oracle, etc.).
   - Execute the selected DDL script to create the required report tables in the database of your choice.

Note: We have currently provided DDL scripts for SQL Server, MySQL, and Oracle. If you require creating the report tables in a different database, we kindly request you to use these DDLs as a reference and make the necessary modifications to suit your target system.# Importing VQL Files

Importing VQL Files:

Within the provided driver package, you will find two distinct VQL files required for the setup. Please follow the steps below to complete the import process:

1. **Download and Extract Files**
   Begin by downloading the **ViewCompare_vql_export_report_views.zip** file. Once downloaded, extract its contents to the folder in the Host server.

2. **Update the Properties File**
   Locate the **viewcompare_vql_export_report_views.properties** file and update it with the appropriate configuration values:

   - Modify the parameters **<JDBC_URI>, <USERNAME>, <CATALOGNAME> and <SCHEMANAME>** so that they correctly point to the data source where the report tables have been created.

   - Encrypt the source connectivity password using the appropriate script:
     - On Windows: **DENODO_HOME/bin/encrypt.bat**
     - On Linux: **DENODO_HOME/bin/encrypt.sh**

- After encryption, replace the **<PASSWORD>** parameter in the properties file with the encrypted value.

3. **Import the VQL:**
   - Import the **Viewcompare_vql_export_report_views_<target_rdbms_database>.vql** file along with the updated properties file to ensure the configuration is applied correctly.
   - Finally, download and import the **viewcompare_vql_export.zip** file. During the import process, use the password **admin** when prompted.

Setup for Other Databases

Note: If the report tables were created in a database other than **MSSQL**, **Oracle**, or **MySQL**, please follow the steps below to ensure proper integration with the framework.

Steps to Follow

1. **Create a Datasource Connection**
   - Set up a new datasource connection in the `vdb_viewcompare` VDB.
   - This connection should point to the database where the report tables were created.
2. **Create Base Views**
   - Once the connection is established, create base views on top of the report tables created in the previous step.
   - Ensure that the base views strictly follow the required naming standards:

Table Name	Base View Name
metadata_comparison	bv_metadata_comparison
row_count_comparison	bv_row_count_comparison
checksum_comparison	bv_checksum_comparison

Important

Following the above naming conventions is mandatory. This ensures that the framework can correctly identify and utilize the views without any issues.

Data Source Configuration

Please configure the following data sources to ensure proper connectivity with the respective environments:

1. **Source Denodo Server Connection**
   - **ds_source_denodo_server**: Configure this data source to establish a connection with the **Source VDP server**
   - Ensure that all connection parameters (such as host, port, database name, and credentials) accurately reflect the **Source VDP server**.
   - Provide appropriate values for the aforementioned parameters, as outlined below:
     - **Host** – Specify the hostname of the Source VDP server.
     - **Port** – Specify the port number on which the Source VDP server is running.
     - **Database Name** – Specify the name of the Virtual Database used to establish the connection.

       The above parameters must be configured within the JDBC URI.

     - **Credentials** - Provide valid username and password details in the user configuration to authenticate and access the Source VDP server.

2. **Target Denodo Server Connection**
   - **ds_target_denodo_server**: Configure this data source to connect to the **Target VDP server**.
   - Ensure that all connection parameters (such as host, port, database name, and credentials) accurately reflect the **Target VDP server**.
   - Provide appropriate values for the aforementioned parameters, as outlined below:
     - **Host** – Specify the hostname of the Target VDP server.
     - **Port** – Specify the port number on which the Target VDP server is running.
     - **Database Name** – Specify the name of the Virtual Database used to establish the connection.

       The above parameters must be configured within the JDBC URI.

   - **Credentials** - Provide valid username and password details in the user configuration to authenticate and access the Target VDP server.

3. **SSH Script Execution Configuration**
   - The **ds_ssh_script_executor** should be configured with the required connection details to establish an SSH session with the **Host VDP Server** where the Python script will be executed.
   - This configuration enables secure remote execution and should include the following parameters:
     - **Hostname or IP Address** of the target machine
     - **SSH Port** (default is 22, unless customized)
     - **Username** with sufficient privileges to execute the script
     - **Password** (or relevant authentication method, if applicable)

   Ensure that the **Host VDP Server** is reachable over the network and that SSH access is properly enabled and permitted for the provided credentials.

Python Script Setup and Configuration

A Python Script is used to determine the checksum of each record of the **Source VDP server and Target VDP server** and perform the row level data comparison**.** Please follow the steps below to configure the Python script.

1. **Copy the Python Script**
   Copy the **checksum_py_script.py** file to a desired directory on the **Host VDP Server** where it will be executed. Ensure that the desired directory has the necessary permissions for script execution.

2. **Update Connection Parameters**
   Open the Python script and update the following parameters to establish JDBC connectivity with the Denodo Virtual DataPort server (Host VDP server):

   - JDBC_DRIVER
   - JDBC_URL
   - USERNAME
   - PASSWORD

These parameters should be configured with the appropriate values corresponding to the Denodo environment to ensure successful connectivity.

3. **Configure JDBC Driver Path**
   - The **JAR_PATH** parameter specifies the location of the JDBC driver required for Denodo connectivity.
   - Copy the **denodo-vdp-jdbcdriver.jar** file from the following directory:**DENODO_HOME/lib/extensions/jdbc-drivers/vdp-9/denodo-vdp-jdbcdriver.jar**
   - Place this **denodo-vdp-jdbcdriver.jar** file in the same directory as the Python script (or in another accessible location).

4. **Update JAR Path in Script**
   After placing the JDBC driver, update the **<JAR_PATH>** parameter in the Python script to point to the directory where the JAR file is located.

Configuring the Python Script Location in Design Studio

To ensure that the Python script can be executed correctly, its location must be configured within the corresponding base view in Design Studio. Please follow the steps below:

1. In Design Studio, navigate to the base view named **bv_ssh_python_script_executor.**
2. Open the view and go to **Edit → Source Refresh**.
3. Provide the full path to the directory where the **checksum_py_script.py** file has been placed.

Ensure that the specified path is accessible from the **Host VDP server**.

Executing Your Tests

The Execution steps will outline details about each validation we do as part of ViewCompare. Follow the steps below, which include specific instructions on utilizing the relevant PL/VQL stored procedures for each check.

The below table summarize the syntax and results expected from the tests

Test Name	**PL VQL Procedure /Final Executable Syntax**	**Parameters**	**Result View**	**Decoding the Results**
Metadata Comparison	metadata_check *(source_dbname, source_view_name, target_dbname, target_view_name)*	*source_dbname*(name of the source database) *source_view_name*(name of the source view) *target_dbname*(name of the target database) *target_view_name(*Name of the target view)	**bv_metadata_comparison**	Refer to Appendix A
Count Check	count_check(source_dbname, source_view_name, target_dbname, target_view_name)	source_dbname(Name of the source database) source_view_name(Name of the source view) target_dbname(Name of the target database) target_view_name(Name of the target view)	**bv_row_count_comparison**	Refer to Appendix B
Check Sum Comparison	SELECT * FROM bv_ssh_python_script_executor WHERE source_dbname, source_view_name, target_dbname, target_view_name, source_primary_key_fi	source_dbname(Name of the source database) source_view_name(Name of the source view) target_dbname(Name of the target database) target_view_name(Name of the target view)	**bv_checksum_comparison**	Refer to Appendix C

	eld, target_primary_key_field	source_primary_key_field(Primary key field in the source view) target_primary_key_field(Primary key field in the target view)		

Appendix A:

Summary of Observations with Explanations:

CATEGORY	DETAILS
datatype or precision	Mismatches in column_vdp_type, column_jdbc_type, column_size, etc.
type decimal, column radix, primary key definition, null value definition, field description, autoincrement definition	All these values are self explanatory and if you need additional info, you could refer to the GET_VIEW_COLUMNS documentation.

source_view_name	target_view_name	column_name	source_datatype	target_datatype	observation
bv_employees	bv_employees	department	NVARCHAR	NVARCHAR	mismatch in "field description"
bv_employees	bv_employees	firstname	NVARCHAR	NVARCHAR	mismatch in "datatype or precision" "null value definition"
bv_employees	bv_employees	lastname	NVARCHAR	TIMESTAMP	mismatch in "datatype or precision" "null value definition"

sample output for bv_metadata_comparison view

Appendix B:

Summary of Observations with Explanations:

RESULT TYPE	DESCRIPTION
Matching	Row counts are equal in both views
Mismatch	Row counts differ between source and target

Execution Steps

- Execute the PL/VQL procedure - **count_check** from the **Host VDP Server** with the required input parameters.
- Results will be available in the view **bv_row_count_comparison.**

source_view_name	target_view_name	source_row_count	target_row_count	observation
bv_employees	bv_employees	6	6	matching

sample output for bv_row_count_comparison view

Appendix C:

Observations:

OBSERVATION TYPE	DESCRIPTION
Missing in Source	Records present in target but not in source
Missing in Target	Records present in source but not in target

Mismatch	Records exist in both but data differs

source_view_name	target_view_name	primary_key	observation
bv_employees	bv_employees	7	missing in source
bv_employees	bv_employees	6	missing in target
bv_employees	bv_employees	3	mismatch

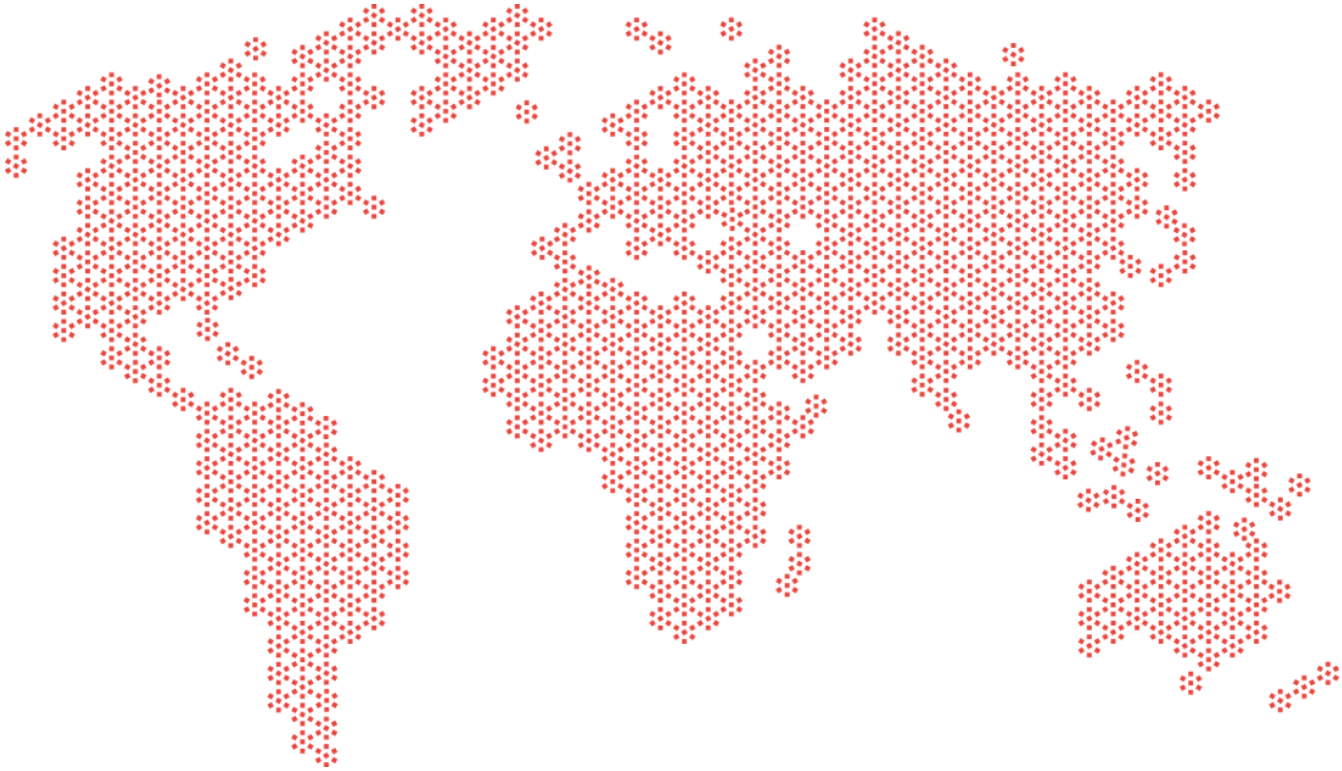
sample output for bv_checksum_comparison view

Sample execution timings for the Check Sum comparison:

Number of Rows	Number of Fields	Time taken
100000	100	40 seconds
500000	100	120 seconds
1000000	100	800 seconds

Limitations and Important Considerations

- Currently the framework accepts input as a single view (source and target each) for comparison. In order to preserve the server overhead, the framework executes it for one view at a time for all the checks.
- For Count Check and Metadata Comparison, in case the virtual database is provided incorrectly, the stored procedure will fail stating the virtual database is missing. In case the view names are incorrect, it is handled as an exception and the output would record the same in the respective output views.
- For Metadata Comparison, if there are multiple mismatches for a single column, it will be shown as multiple messages enclosed in double quotes. For example, **mismatch in "datatype or precision" "null value definition".**



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